Expressions

So far in the course we have learned many techniques for creating motion graphics. We've learned how to keyframe still images, work with videos and even track points in space within a video. Another method for creating motion to use a programming language to assist us. In this lab we will learn the basics of expressions in Adobe After Effects. This will give us an understanding of what an expression is, and how to use it to animate objects in our project.

Setup

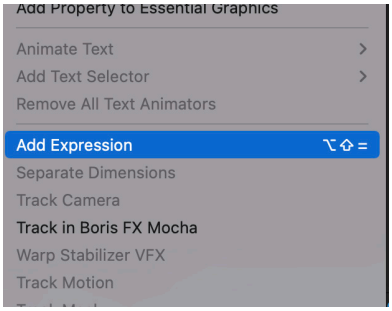
Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

https://youtu.be/vBEMlnRgR8U

Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *Lab-Expressions.zip*. Make sure to unzip the folder completely to use the files inside appropiately. Open *expressions.aep* into Adobe After Effects.

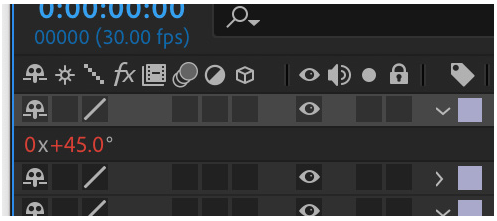
Expressions

Expressions are essentially using programming code to animate objects within our project. In the case of Adobe After Effects we can use a programming language called Javascript to assist us. This lab is not going to teach you all the ins and outs of learning Javascript or even programming concepts itself, but will hopefully give you an idea of how to expressions are used in a motion graphics environment, and how we can utilize them to create interesting effects.



In the composition you'll see stars and three spaceships. We will begin with the *red spaceship* layer. Begin by getting the rotation layer property for the red spaceship layer by hitting the letter *'R'*. We've animated and created keyframes using the timeline before, but now we will learn how to apply expressions to a property.

With the rotation property selected go to *Animation > Add Expression*. This will open a new area in the timeline that looks like a code editor. You can also get this area by *option* clicking on the property in the timeline as well. You can tell when a property is using expressions because it's values should be in red.

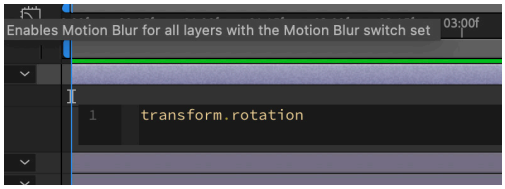


You'll see the following expression:

```
transform.rotation
```

This code represents the current rotation value of *red spaceship* layer. At the moment it has not be set. If you remember back to math class and basic algebra you may remember if you had an expression x =

2, you are stating the variable of *"x"* is equal to the value of *"2."* In the same vein, the *transform.rotation* is the rotation property, but it does not have a value currently. We could use an equal sign like in math such as this:



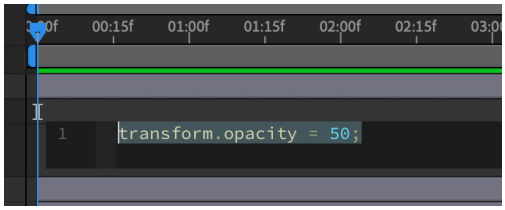

```
transform.rotation = 45;
```

In other words, the rotation property of this layer is equal to, or set to, 45 degrees.

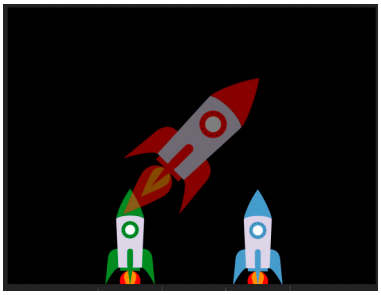


Set Transparency

Now that we wrote some code that set the rotation, let's write code that will set the transparency of a layer in After Effects. Select the *red spaceship* layer again and select the opacity property or press the letter *"T"*.



With the opacity property highlighted go to *Animation > Add Expression* to once again open the code editor. This time we will use code to set the transparency fo this layer. Write the following code:

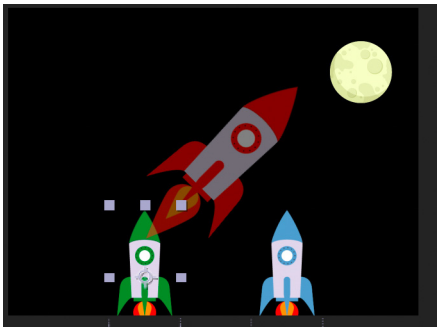


```
transform.opacity = 50;
```

This code is saying that opacity of this layer is equal to *50* or *50%*. Your composition should look roughly like the image on the right.

Set Position

Let's write code to set the position of the *moon* layer. Select the position property or press the letter *'P'*. With the position property highlighted go to *Animation > Add Expression* to once again open the code editor. This time we will use code to set the position of the moon layer. Write the following code:



```
transform.position = [550, 100];
```

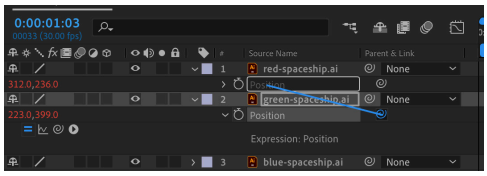
The following code sets the x and y position of the layer in the brackets. The moon should now be in the top right hand corner.

Properties to Affect Other Properties

In addition to writing code to affect a layer we can use expressions to have the value of a property set by another layer. In this example we use the position value of the *red spaceship* to match that of the *green spaceship*.

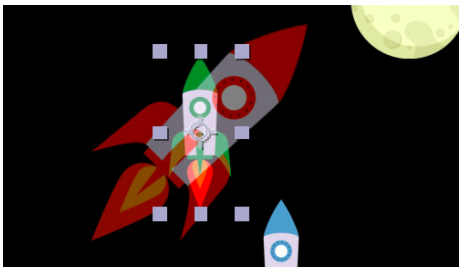
Begin by selecting both the red and green spaceships and open both position properties by pressing the letter *'P'*.

Select the *green spaceship* layer and add an expression to the position property. In the timeline grab the *pickwhip* and drag it to the position property of the *red spaceship* layer. You can view the image on the right as a reference. The result of this is the following code:

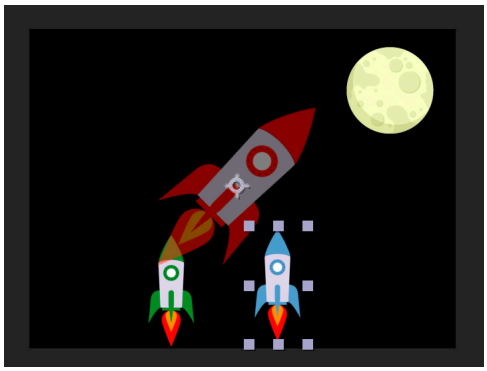



```
thisComp.layer("red-spaceship.ai").transform.
position;
```

What this code is doing is the green spaceship's position is set to the position of the red spaceship. The result will be the green spaceship moves its position.



Repeat the process with the blue spaceship. Highlight the position property of the blue spaceship and add an expression. Use the pickwhip to set the expression equal to the position of the red spaceship layer. The result will be that both the blue and green spaceships are in the same position as the red spaceship.

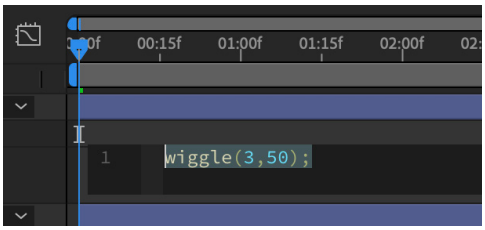


Highlight the red spaceship layer and adjust the position property, you'll notice that the blue and green spaceships move with it. At the moment the positions of the blue and green spaceships are overlapping the red spaecship, which is a bit akward. Use the *anchor* property for the blue and green spaceship to adjust the spacing. Plae the green spaceship on the lower left of the red spaceship and the blue spaceship on the lower right. Use the image on the right as a reference.

In the timeline set the time marker to the beginning. Move the red spaceship to be at the bottom center of the screen. Keyframe the position to move the red spaceship to the top of the screen after 3 seconds. Play back the results and all three spaceships will move together!

Wiggle

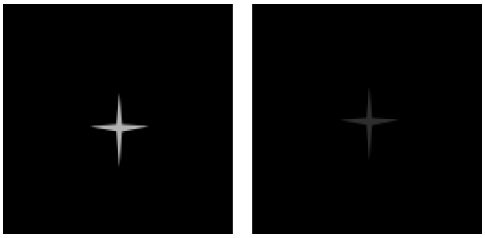
In our composition we have a group of stars in the star composition. Let's say that we want to have the stars 'twinkle'. Twinking is really just the opacity of the star changing from high-to-low-to hight many times. We could create keyframes to create that animation, but it would actually be easier to use expressions.



Open the *Stars* composition and select *Star1*. Open the opacity property by pressing the *"T"* key. Delete the existing code and type the following:

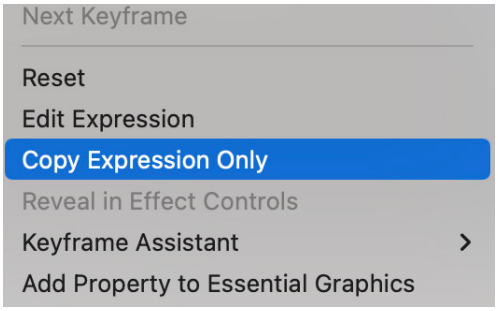
```
wiggle(3, 50);
```

What the wiggle expression does is randomly change the value from its current value by a certain amount. In the case above it will affect the opacity's value by 50% each time it is re-evaluated. How many



times this occurs per second is set by the first value, which is in this case *'3,'* In other words, the opacity of the star should change three times per second, up to 50% per time. Scrub through the timeline and you should see the value of the opacity change many times.

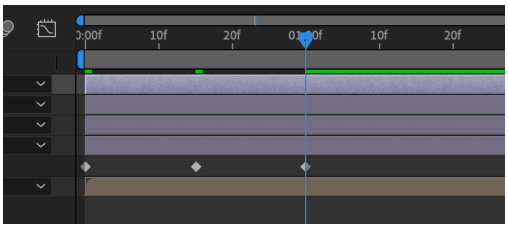
Copy an Expression



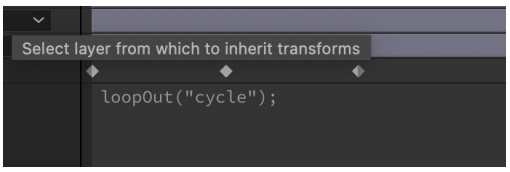
Now that we've added the wiggle expression to *Star1*, let's add it to the other stars as well. We could write the expression again for each layer, but it would easier and faster to simply copy and paste the expression between layers.

Turn on the remaining star layer visibility toggles. Select *Star1* and right click, in the pop up dialogue box choose the option *"Copy Expression Only."* Select the remaining stars and go to *Edit > Paste* to paste the wiggle expression to each layer. Scrub through and all of the stars should be twinkling.

Cycle Keyframes



Many times in a project we may want to repeat an action, such as as object moving up and down, or a layer pulsing larger and smaller. We could simply repeat the action with many keyframes, but there is an easier way to do this with expressions.



In the Expressions composition place the time marker at the beginning. Select the moon layer and set the scale of the layer to *50%* and set a keyframe. Move to the frame 15 and set the scale to *70%*. Once again move to the 1 second point and set the scale back to *50%*. The result should be a short animation of a pulse, or the moon growing and shrinking. We want to repeat this action so it continues to pulse for the remainder of the composition.



Add an expression to the scale property of the moon. Remove any existing code. Add the following:

```
loopOut("cycle");
```

This code tells After Effects to loop or repeat the keyframes existing in the layer for the remainder of the composition.

*E*XPORT

Great job! Now let's export the final project. With the *Final* composition open in the timeline go to *Composition > Add to Render Queue*.

In the *Render Queue* look for *Comp Name* make sure it is set to *Final*.

Under *Render Settings* make sure it is set to *"Best Settings".*

Under *Output Module* make sure it is set to *"Lossless"* make sure the format is set to either Quicktime or AVI.

Finall, use the *Output To* option save the animation to your Desktop with the label *lastname-post-production.*

SUBMISSION

Upload your exported project video to your YouTube account. Publish the video and you will copy and paste the video URL into a text document to be submitted.

In addition to the final video url please upload the *lastname-expressions.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.

Put the URL document and *.aep* into a folder, compress it and upload it to Canvas for grading.